

On the Value of Sales Quotas

Nathan Gardner Hattersley

Latest Revision: September 25, 2026

([NEWEST VERSION AVAILABLE HERE](#))

Abstract

State franchise laws preclude automobile manufacturers from most classical vertical restraints, including resale price maintenance, quantity forcing, and exclusive territories on the manufacturer's terms, and they are widely blamed for the pricing inefficiency of new-car retailing. I argue this critique overlooks the role of the allocation process, in which manufacturers reward dealers who meet quarterly sales targets with expanded trim assortments. Using the universe of Texas new-vehicle registrations from 2017–2019 and the Ford franchise network, I develop a dynamic model in which dealers set prices in anticipation of assortment rewards. I recover latent compliance with sales targets from observed assortments via a Monte Carlo EM estimator, exploiting a nonparametric identification result for finite mixtures, and recover marginal costs using the moment-inequality estimator of Bajari, Benkard, and Levin (2007). Roughly 90% of dealer-quarters meet their target, and dealers price on average **XX%** below the static-pricing benchmark because of the dynamic incentive. Allocations act as a quantitatively important vertical restraint despite their ambiguous legal standing.

New car retailing in the United States is subject to a patchwork of state-level regulations that strongly restrict the ways in which manufacturers may interact with their franchised dealers. Chief among them is the requirement that manufacturers not vertically integrate forward into retail; manufacturers must maintain an arm’s length relationship with independent dealers that form a traditional-format franchising network. Furthermore, these franchising relationships are precluded from several standard vertical arrangements upstream firms might use to discipline the vertical structure and correct distortions from double marginalization. In Texas, the state that I study here, manufacturers must charge uniform wholesale prices to their dealers; they may not engage in retail price maintenance; they may not engage in quantity forcing; they must uphold territorial protections for existing dealers;¹ and they are broadly enjoined from behavior which “[discriminates] against a dealer [...] on the basis of a formula intended to gauge sales performance.” (See Tx. Occ. Code §2301 for the full list.)

This regulatory context is broadly maligned by economists (c.f. Smith 1982 and Lafontaine and Scott Morton 2010) who rightly see these protections for dealers as creating an environment where they are able to extract a protected rent without the threat of market forces to compete profits down to a more socially efficient level. In this paper, I argue that this critique ignores the important role that the allocation process plays as a vertical restraint. Allocation is the process of determining *which* particular finished cars are distributed to *which* dealerships. An important way this allocation is decided is by manufacturer-set sales targets. Each dealer will receive their “base” allocation, which are cars that they receive regardless of their sales performance. The (rolling quarterly) sales target is part of an incentive mechanism designed to lower retail prices and increase output: If the dealer meets their manufacturer-set sales target, then they are eligible for a “bonus” allocation, which typically includes more desirable models and trims² either not included or under-provisioned in the standard allocation. This practice appears to violate the letter of the law, in particular the broad anti-discrimination clause in Tx. Occ. §2301, yet this practice is widespread and appears to be settled case law.³ Despite an ambiguous legal standing, allocations in practice play an important role in mediating the competitive effects of state-level restrictions, one which appears not to have received sufficient attention in the academic literature.

¹New dealerships must meet a long checklist of requirements to justify entry as “necessary,” effectively creating and protecting exclusive territory for incumbent dealers.

²A trim is a variation of a model. There is typically a “base” trim of a model and other trims roughly vertically differentiate themselves from the base trim. For example, the Toyota Corolla has a “standard” trim and a “sport” trim.

³See *Hyundai Motor America v. New World Car Nissan, Inc. et al.*, 581 S.W.3d 831 (Tex. App. 2019), an appeals court ruling based on a complaint originated via the DMV’s internal process contesting World Car Hyundai’s allocations. An administrative law judge and the appeals court both upheld Hyundai’s allocation process as legal, which is broadly the same process across all manufacturers. A public records request and review of internal DMV complaints over a ten-year period returned no other matters pertaining to allocations, implying this is widely viewed as settled law.

Studying Ford dealers in particular, I develop and estimate a model that captures the effect of this sales incentive mechanism. This model broadly follows the description of the allocation system that I gave above. It also allows for tractable estimation in a world where the size of dealers’ product assortments (conditional on local market conditions) is informative for whether or not they met their sales target. My game sees dealers set prices dynamically with the anticipation of expanded assortment as a reward in future periods. I use observed assortments and quantity sold to recover the sales target rule and a latent indicator of whether each dealer met their sales target. Next, I use the dealers’ observed pricing behavior to recover marginal costs via dynamic estimation à la Bajari, Benkard, and Levin 2007, and with these estimates I am able to compare observed prices to counterfactual prices that would arise in static equilibrium without incentives. I find that manufacturers broadly set achievable sales targets: Across all dealer-quarter combinations, roughly 80% meet their sales target. I also show that despite high compliance, the dynamic incentives introduced by these sales targets have material impacts on dealers’ pricing behavior. Compared to a static pricing benchmark, I estimate that dealers on average price their cars XX% lower because of the dynamic incentive created by the allocation process. These results provide important context to the debate over behavioral restrictions in car retailing, and they also serve as an important benchmark for the industrial organization literature on vertical restraints and their effectiveness in eliminating the double marginalization externality.

0.1 Related Literature

This paper fits into a literature studying the impact of vertical restraints. As far back as Rey and Tirole 1986 and Tirole 1988, economists realized that some measure of price or quantity controls between two vertically related parties can assuage the effects of double marginalization. Double marginalization is a phenomenon arising when an upstream and a downstream firm both have market power. Each marks up its output price over marginal cost (“taking double the margin”), and prices are ultimately higher and output lower than if the industry had been controlled by one vertically integrated firm. See Lee, Whinston, and Yurukoglu 2021 for a comprehensive review.

The theory of vertical restraints developed as a way to explain the types of contractual relationships observed in practice, and it demonstrates how simple provisions can rectify the double marginalization externality. As discussed in Rey and Tirole 1986, Tirole 1988, and many others, quantity forcing or resale price maintenance can require the downstream firm to act according to the upstream’s wishes, leading to prices and output that correspond to the vertically integrated prices and output in simple models. As mentioned, Texas law expressly prohibits quantity forcing and resale price maintenance. Dealers are also insulated from threats of entry, to make matters worse, perpetuating pricing power that could typically be ameliorated by competition

from newcomers. Among other concerns about the franchising relationship,⁴ manufacturers appear unable to mitigate the issues of double marginalization. However, the literature appears to have not sufficiently appreciated the way in which allocations can act as a vertical restraint.

A handful of empirical studies have addressed other aspects of the car franchising relationship. Nurski and Verboven 2016 discuss an aspect present in Europe but not in the United States; that is, the ability for manufacturers to enforce exclusive dealing contracts. They find dealers have an obvious incentive to unilaterally deviate and sell additional makes, but the franchise network as a whole is more profitable under exclusive dealing. Albuquerque and Bronnenberg 2012 address the impact of exit restrictions on dealer networks during the Global Financial Crisis. They use transaction data from the San Diego metro area to calculate the operating costs and margins at area dealerships and find that in this case, General Motors would have benefited from the ability to terminate one of their local dealers' contracts. Biesebroeck and Verboven 2024 gives a relevant review of a broader literature surrounding automobile retailing.

One closely related paper is Conlon and Mortimer 2021. They study the phenomenon of all-units discounts (AUDs) in vending machine sales—an arrangement whereby vending machine operators that meet a sales target set by the manufacturer receive a discount on all units sold (retroactively) in a given period. They partner with a vendor subject to an AUD from the Mars candy bar company. They recover an estimate of the efficiency gains from the sales contract, and they find that the AUD plays an important role in Mars inducing behavior from vendors that looks more like that of a vertically integrated firm. However, they also find foreclosure effects that outweigh the consumer surplus gains from vertically aligned incentives, as in order to meet the sales target, the vending machine operator is more likely to stock the lower-selling candy from Mars rather than better-selling candy from its competitor, Hershey.

There are several important differences between Conlon and Mortimer 2021's work and mine here. First, car dealers exist in franchising relationships that are usually exclusive—moreover, they are not making assortment decisions on the fly between two manufacturers, and there is no scope for foreclosure effects in the same sense as in vending machines. Second, prices are fixed *ex ante* in vending machines, and the main decision variables from the operator's perspective are how to stock limited shelf space in the machine and how frequently to send employees to restock the machines. Hence, the foreclosure effects from opting for Raisinets over Reese's Cups is salient, and there is little scope for vertically aligned incentives to increase consumer surplus otherwise. Car dealers set prices on a per-car basis, so there is ample leeway for an adequately

⁴The concern being franchising “effort” in the sense of facility investments, sales effort, etc. See Lafontaine and Scott Morton 2010 for the automobile industry and Lafontaine and M. E. Slade 1998 for a more general description of franchising as a principal-agent problem. See also Murry 2017 for a treatment of the externality in advertising generated by the dealership system.

motivated dealer to engender surplus gains. In that sense, my setting is significantly more favorable for finding positive effects from sales target mechanisms.

Worth noting also is a long literature on the effects of sales targets when included as part of a worker’s compensation package. These papers tend to focus on how nonlinear contracts (with a bonus structure for meeting sales targets) tend to induce higher worker effort. For example, Misra and Nair 2011 develop and estimate a dynamic structural model of salesperson effort using data from a contact lens manufacturer. There are even instances, such as in Jindal and Newberry 2022, where the model includes the salesperson’s ability to set prices, and there is discussion in Lafontaine and M. Slade 2007 on how this principal-agent problem of worker compensation interplays with the notion of firm boundaries and how firms may think about the decision to integrate forward into retailing. However, this literature focuses primarily on the “effort” effects of varying compensation schemes, where risk aversion, limited liability, and the insurance effect of wages play a primary role in both rationalizing why these schemes are favored and in explaining worker incentives. The model that I develop of the allocation process is similar in spirit to these principal-agent models, however the goal in my model is simply to investigate how a sales target mechanism between two risk-neutral businesses transacting at arm’s length can minimize distortions from double marginalization.

The rest of the paper is as follows: In the next section, I provide a longer explanation of the relevant context surrounding car retailing, and I present a basic theory model capturing the important aspects of allocations and demonstrating where efficiency gains lie. Then I present the data I use for estimation, show reduced-form patterns that align with the predictions of my model, and I develop and estimate a structural model of consumer demand for cars. I develop a dynamic structural model of dealer allocations and pricing behavior, then show how to identify and estimate its primitives. Finally, I present results, discuss interpretations of counterfactuals, and a final section concludes.

1 Background and Theoretical Model

1.1 Automobile Franchising Context

As mentioned at the outset, a unique regulatory history has led to a patchwork of state laws that heavily constrain the ways in which manufacturers interact with their franchise dealers. These franchise laws emerged early in the history of automotive sales, as smaller franchisees were subject to the whims of more dominant manufacturers and abided by borderline-extortionary terms of trade (Lafontaine and Scott Morton 2010).

Over time, state legislation accumulated in pursuit of righting this perceived wrong and protecting dealers’ franchise claims.

In Texas, Occupations Code §2301 regulates the interaction between dealerships and their manufacturers. Among provisions laid out in this chapter are bans on quantity forcing, price discrimination, reduced threat of entry, reduced threat of franchise termination, and a ban on direct manufacturer competition. There is also a clause forbidding “inequitable treatment” of two franchise dealers “as a result of the application of a formula or other computation or process intended to gauge the performance of a dealership. . . .”

The clause in Texas Occupations Code §2301 forbidding inequitable treatment of dealers specifically mentions formulas that measure dealer performance. One of the core mechanisms of the allocation process is a sales target that manufacturers set, with extra trims and better trims used as a reward for meeting the target. Dealers in interviews describe a process that they have little control over and little insight into. One Ford dealer said that they get assigned their sales target, and they know that if they meet it they get a better allocation, but that is generally the extent of their understanding. There is only one lawsuit about allocations that has made it to public court in Texas—that is the case of *Hyundai Motor America v New World Car Nissan Inc*, originally brought in 2013. A Hyundai dealer in San Antonio did not participate in a manufacturer-sponsored program to improve facilities. As a result, Hyundai gave them an allocation that made them mathematically unable to meet their sales target. An Administrative Law Judge ruled in favor of Hyundai. While the case is only tangentially about the sales target mechanism itself, the judge found nothing about what Hyundai did to be illegal. I made a records request and reviewed all of the formal dealer complaints filed with the Department of Motor Vehicles over the last ten years. There were no complaints about a dealer’s allocation or sales targets. Hence, this appears to be a settled matter in a practical sense, even if it appears to violate the letter of the law.

The toy model and supply-side model I estimate will demonstrate in a more formal sense what this allocation-sales target mechanism is. It is important to note that even if dealers feel they have little control over the process, this cannot be interpreted as quantity forcing. This practice is illegal and does not appear to be loosely enforced like sales targets are. Hence, my model uses incentive compatibility to induce dealers to lower prices and increase sales, with the reward being an expanded product assortment. The easiest modeling choice for me was for the manufacturer to choose the product set $\mathcal{J}$. This is tractable with standard industrial organization methods and avoids wavering too close to a model where manufacturers explicitly set quantities in some fashion—in effect modeling an illegal behavior. I will demonstrate why product availability is a useful approximation of reality in Section 2.

1.2 A Simple Model

In this section I present a toy model that highlights the basic economic forces behind my allocations model, and it shows how allocations may reduce efficiency loss from double marginalization. An upstream firm sets a sales target on a low-quality good and the downstream subsequently sets prices, with availability of the high-quality good contingent on meeting the sales target.

The two notable differences with my empirical model are that this toy model is static and that it allows the upstream firm to adjust wholesale prices freely. The static context was chosen because my main results are easier to prove here than in a repeated game setup. It should be apparent how the results can translate to the dynamic empirical setting. Letting the upstream firm adjust wholesale prices is a modeling choice that I do not make in my empirical model for tractability reasons, but as I will show this means my empirical results are a more conservative estimate of efficiency gains.

Imagine a monopolist manufacturer that produces two vertically differentiated goods, and a monopolist retailer that sells them to consumers. Consumers have the demand function similar to Tirole 1988 §2.1.1. Consumer types are indexed by scalar θ that is distributed on a compact interval $[\theta, \bar{\theta}]$ according to continuously differentiable CDF F . Their respective utilities for the low-quality good L and high-quality good H are

$$u_L = \theta v_L - p_L$$

$$u_H = \theta v_H - p_H$$

With $v_H > v_L$. Assuming both goods are sold, demand is defined by two cutoff points

$$\Delta_H = \frac{p_H - p_L}{v_H - v_L}$$

$$\Delta_L = \frac{p_L}{v_L}$$

corresponding respectively to the type of the consumer that is just indifferent between the two goods, and to the type of the consumer that is just indifferent between the low-quality good and the outside option not to consume. The retailer faces demand function

$$D_L(p) = F(\Delta_H) - F(\Delta_L)$$

$$D_H(p) = 1 - F(\Delta_H)$$

with fraction $F(\Delta_L)$ choosing the outside option. Assume that the manufacturer faces increasing marginal costs of production, in this case a simple additively separable quadratic

$$C(q_L, q_H) = \frac{1}{2}c_L q_L^2 + \frac{1}{2}c_H q_H^2$$

Assume zero costs of retailing. Now consider the following extensive form game between the manufacturer and the retailer: The manufacturer moves first and sets wholesale prices (w_L, w_H) as well as a sales target φ . The retailer then sells for two periods. In the first period, the assortment available to them to sell is $\{L, H\}$. They choose prices $\{p_L^{(1)}, p_H^{(1)}\}$ and realize demand $q^{(1)}$.

In the second period, the assortment available to the retailer to sell depends on the sales target and their sales in the first period. If total sales exceed the sales target, that is,

$$q_L^{(1)} + q_H^{(1)} = 1 - F(\Delta_L^{(1)}) \geq \varphi$$

then the retailer will have $\{L, H\}$ available to sell once again. However, if they do not meet their sales target, then they will only have $\{L\}$ available to sell. The goal of the retailer given a manufacturer's wholesale price and sales target is to maximize total profit over the two periods:

$$\begin{aligned} \pi^R &= \max\{\pi^{RC}, \pi^{RD}\} \\ \pi^{RC} &= \max_{p^{(1)}, p^{(2)}} (p^{(1)} - w) \cdot q^{(1)} + (p^{(2)} - w) \cdot q^{(2)} \quad \text{s.t.} \quad q_L^{(1)} + q_H^{(1)} \geq \varphi \\ \pi^{RD} &= \max_{p^{(1)}, p_L^{(2)}} (p^{(1)} - w) \cdot q^{(1)} + (p_L^{(2)} - w_L) \cdot q_L^{(2)} \quad \text{s.t.} \quad q_L^{(1)} + q_H^{(1)} < \varphi \end{aligned}$$

I let π^{RC} be the retailer's profit when they comply (that is, meet their sales target) and π^{RD} be their profit when they defect (do not meet their sales target). I write that the retailer must choose not to meet the sales target if they defect, in order to emphasize that they have a different assortment in the second period. However, the sales target need not be binding: the manufacturer could set $\varphi = 0$, the retailer would be free to set prices as they wish, and we would obtain the double marginalization outcome. In a more realistic scenario, the sales target binds and the retailer must lower their prices below the static profit-maximizing level to meet the target.

The manufacturer’s goal is to set wholesale prices and a sales target that maximize profit subject to a participation constraint:

$$\begin{aligned}\pi^M &= \max_{w, \varphi} (q^{(1)} + q^{(2)}) \cdot w - C(q_L^{(1)}, q_H^{(1)}) - C(q_L^{(2)}, q_H^{(2)}) \\ \text{s.t. } & q_L^{(1)} + q_H^{(1)} \geq \varphi\end{aligned}$$

To illustrate the efficiency gains that a mechanism like this can produce, I calculate price equilibrium under an example parameterization of this game, and I compare it to the benchmarks of vertical integration (first-best) and double marginalization. I set $(c_L, c_H, v_L, v_H) = (2, 2.5, 2.5, 4)$. In order to make the nested optimization problem more tractable by my solver framework, I must introduce a small mean-zero shock (Normal with $\sigma = 0.05$) to period-one demand so that the retailer’s first-order conditions are continuous in φ . (The participation constraint now binds in expectation as well.) Table 1 compares the outcome of the allocation game against the two benchmarks. Note that the manufacturer sets a sales target that is higher than the double marginalization outcome, and the retailer is forced to lower prices in order to meet it. At the same time, because the sales target is binding in expectation, the manufacturer is able to raise wholesale prices beyond double marginalization levels in order to extract rent from the retailer. Ultimately, both producer and consumer surplus are higher overall, and we are able to recover a large fraction of the lost surplus from the double marginalization externality using this mechanism.

Table 1: Outcomes from Simple Model

	VI	DM	Allocation
(q_L, q_H)	(0.147, 0.260)	(0.067, 0.109)	(0.050, 0.220)
(p_L, p_H)	(3.231, 4.941)	(5.474, 8.464)	(4.947, 6.118)
(w_L, w_H)	—	(2.881, 4.415)	(4.790, 4.565)
π^M	1.656	0.656	1.181
π^R	—	0.616	0.349
Producer Surplus	1.656	1.272	1.530
Consumer Surplus	0.425	0.130	0.407
Total Surplus	2.081	1.402	1.937
Frac. Recovered			0.788
φ			0.246

There are several differences worth noting between this simple model and the empirical model that I will implement. First, the payoff to the retailer for meeting their sales target here is that they are able to statically profit-maximize in the second period. My empirical framework is that of a repeated game, so the intuition of a dynamic incentive to lower prices now for a higher future payoff carries over, but in a slightly different sense. Second, my demand model of vertically differentiated goods meant that the objective of

the manufacturer setting the sales target is essentially to expand the market as much as possible while endogenizing the retailer’s response. My empirical demand model will be a standard differentiated goods discrete choice model from the literature, which lends a less clear interpretation to the market-expansion effect of the sales target. Third, I will still consider a monopolistic manufacturer in my empirical model, but the retail sector will be oligopolistic with spatially (and to a small extent vertically) differentiated retailers. Finally, the reward mechanism will be stochastic: Even with a retailer that meets their sales goals, they do not have a guaranteed assortment in the next period, rather their assortment is drawn from a multivariate Bernoulli whose component probabilities are increased by meeting the sales target.

2 Data and Reduced-Form Evidence

Data on car registrations come from the Texas Department of Motor Vehicles. It contains the universe of registrations in Texas from the beginning of 2017 to the end of 2019. Each registration contains the vehicle identification number (VIN) of the vehicle registered, the date the registration is effective, the zip code of the owner, the mileage of the vehicle, the type of registration (individual, business, government, etc.), and if it is sold by a dealership in Texas, the dealership’s general distinguishing number (GDN) and sale price of the vehicle. I decode VINs using the National Highway Traffic and Safety Administration database to obtain make, model, and trim information. New vehicle sales are not explicitly flagged and must be inferred, which I do by assuming that cars sold with less than 250 miles on the odometer and within one year of the car’s model year are new. Appendix Figure ?? justifies this choice, as the density of cars sold drops off significantly before 250 miles and the rate of cars that are surely used (greater than one year old) is increasing in vehicle mileage. I focus only on sales to individuals in my analysis.

In addition to the car registration data, I obtained a list from the Department that contains the licensed car dealers in Texas. I match GDNs between registrations and this list to get the set of franchise Ford dealers in Texas, which I manually review to verify accuracy. I geocode the dealers’ locations using address information provided in the list. I am left with a total of 218 Ford dealers, license numbers to attribute registrations to dealers, and the dealers’ locations.

As I have the zip codes of the registrants as well, I use Census and American Community Survey data to build a demographic profile of the consumer at the most granular level possible. I collect the total population of the zip code from the 2020 Census and the share urban, Black, and Hispanic from ACS data, as well as the median income of the zip code. I also calculate distance between the zip code’s centroid and each dealer at which I see purchases made.

Because prices are negotiated, I see different prices being paid for the same car in my data. A change in a dealer’s average sales price in a quarter could be due to changes in how they mark up their cars, but it could also be due to compositional changes in the types of cars they sell. For that reason, I compute a markup measure that captures the percentage deviation a particular car is sold at, relative to the statewide average price for the same car:

$$\tilde{p}_{ijt} := \frac{p_{ijt} - \bar{p}_{jt}}{\bar{p}_{jt}}$$

where i indexes individual transactions, j indexes a unique model/trim combination, t indexes model year, and $\bar{p}_{jkt}$ is the average price of the year-model-trim combination across all dealers. Within a given period, I calculate and store the median markup at which a dealer sells their cars. Changes in median markups reflect dealers raising or lowering their prices, regardless of their allocation.

Trims as decoded by the NHTSA tend to be fairly granular. There are some trims that appear to be only marginal or unique, with very few sales in my data. I drop any trims that have fewer than 100 unit sales altogether to focus the analysis on trims that reasonably frequently awarded in allocations. Table 2 presents information on the number of different trims I keep for each Ford model. The number of trims ranges from one for the F-250 to ten for the Fusion. I also report total sales for each model, along with a measure of availability to dealers. To create the last two columns, I create a panel of dealers by quarter and use an indicator for each different trim that takes value one when I see the dealer sell said trim in a given quarter. I aggregate up to the trim level and take the mean of the indicator to get the fraction of dealers that had a particular trim when it was available. Then, aggregating up to the model level, I take a (sales-weighted) mean and standard deviation of this availability measure.

Table 2 shows that there is substantial variation in the availability of different trims across dealers in Texas. With the exception of the popular F-150 and F-250 models, dealers tend to have only roughly half of the trims available to sell at a given time. There is also substantial variation within models, as some trims are available much less frequently, suggesting they are awarded only to preferred dealers.

Suggestive evidence that these preferred dealers are those that meet their sales targets is presented in Table 3. I present a series of two-way fixed effect regressions of the form

$$y_{ft} = \beta * x_{ft} + \delta_f + \delta_t + \varepsilon_{ft} \tag{1}$$

Table 2: Trims by Model and Variation in Availability

model	No. Trims	Unit Sales	Avg. $1\{j \in \mathcal{J}\}$	s.d. $1\{j \in \mathcal{J}\}$
F-150	6	155174	89.18%	10.42%
F-250	1	61947	87.95%	—
Explorer	5	46976	59.06%	15.10%
Escape	4	42742	59.34%	6.64%
Mustang	5	28794	50.02%	6.53%
Fusion	10	24922	53.49%	15.76%
F-350	2	24714	66.80%	0.91%
Focus	6	23296	46.78%	14.41%
Edge	5	21616	50.83%	11.34%
Expedition	4	12304	45.99%	11.23%
Fiesta	3	11878	49.82%	13.69%

where I decompose the relationship between markup, unit sales, and trim availability. I also categorize dealers as low-, mid-, and high-performing based on their average total sales volume and disaggregate the effects by these groups.

The first two columns simply establish that the law of demand holds with these metrics. Higher median markups are associated with lower sales within a given quarter. The purpose is to establish that dealers can raise their sales by selling vehicles at a lower markup. Interestingly, the disaggregated effect suggests that mid-performing dealers feel this effect most strongly, with low- and high-performing dealers more likely to face sales that are not as responsive to price changes.

The next two columns establish suggestive evidence that extra trim allocations come as a reward for better sales performance. I look at the effect of total sales volume this period on the number of unique trims sold in the next period. Overall the effect is mildly economically significant, in the sense that the interpretation of the coefficient 0.95 is dealers get roughly one extra trim next period for doubling their sales. However, when I split by dealer performance, I find that this effect is not at all significant for low-performing dealers. Rather, the effect is amplified for mid- and high-performing dealers, who appear to get rewarded more handsomely for increased sales.

The last two columns try to combine the two effects together, regressing median markup this period on the number of unique trims available next period. The regression is not as able to tease out a significant effect as the regression of lagged quantity on number of unique trims, but in the aggregate there is still a statistically significant effect suggesting that dealers who increase their markups by 20 percentage points receive one fewer trim in the next period, on average.

Table 3: Markup, Sales, and Trim Regression

	log(total_units)		unique_trims			
	(1)	(2)	(3)	(4)	(5)	(6)
median_markup	-0.922** (0.312)	-0.444 (0.365)				
median_markup $\times$ is_mid		-1.400* (0.642)				
median_markup $\times$ is_high		-1.059 (0.706)				
log(total_units_lag)			0.950*** (0.262)	-0.306 (0.283)		
log(total_units_lag) $\times$ is_mid				1.747*** (0.451)		
log(total_units_lag) $\times$ is_high				3.862*** (1.097)		
median_markup_lag					-4.519** (1.727)	-2.389 (2.088)
median_markup_lag $\times$ is_mid						-7.121 (4.114)
median_markup_lag $\times$ is_high						-3.172 (4.555)
t Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
dealer_uid Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	3,367	3,367	3,148	3,148	3,148	3,148

To conclude, this regression should not be interpreted as causal evidence that allocations are affecting dealer pricing. It is quite suggestive, as allocations and sales targets are on a rolling quarterly basis and I just presented evidence that increased unit sales and lower pricing are associated with a greater assortment of trims in the next quarter. To more credibly address the question of the allocation system's impact on pricing, next I develop and estimate a structural model and use it to estimate dealer pricing in a world without allocations.

3 Demand

I model consumers as choosing both a particular trim and a particular dealership. Because of this, I can be fairly parsimonious and allow mean utilities δ_{jft} to consist primarily of fixed effects for trims and dealers. Because I have the zip code of the consumer purchasing the vehicle, I treat my sample as a full micro-sample and model consumer indirect utility as

$$u_{zjft} = \delta_{jft} + \mu_{zjft} + \varepsilon_{zjft} \quad (2)$$

Here z indexes zip codes, j the Ford trims I treat as inside goods, f indexes dealerships (or “firms”), and t indexes markets, which in my case are quarters. Hence, in a given quarter, consumers can choose from any dealership and choose any Ford trim that said dealership carries.

Mean utility is composed of fixed effects for trim, dealership, and quarter, as well as a mean price effect:

$$\delta_{jft} = \bar{\alpha}p_{jft} + \xi_j + \xi_f + \xi_{q(t)} + \xi_{y(t)} + \gamma \log(J_t) + \Delta\xi_{jft} \quad (3)$$

I choose not to use a fully saturated fixed effect for quarter t , as my counterfactual scenarios will involve nontrivial changes in the size of the choice set. Instead, I use a fixed effect for both the quarter-of-year, $q(t)$, and year, $y(t)$. This allows me to include a penalty on the size of the choice set, J_t , à la Akerberg and Rysman 2005.

The price p_{jft} is the average price at which I observe dealer f selling trim j in quarter t . Even if price dynamics are variable within a quarter—that is, a dealer starts dramatically cutting price near the end of a month or quarter to meet sales targets—I expect average price to reflect enough of the dealer’s pricing behavior to be sufficient for my application. As is standard in the demand estimation literature, I include an unobserved demand shock $\Delta\xi_{jft}$ that creates price endogeneity, for which I will instrument using Hausman instruments.

The zip-code-specific utility μ_{zjft} is composed of interactions between a zip code’s demographic variables D_z and a set of indicators x_j for the car’s vehicle type: sedan, coupe, truck, or SUV. This allows each vehicle type to have differential appeal to each zip code. The set of demographics that I include are the share urban as reported by the ACS, share Black, share Hispanic, and the reciprocal of the median income.

I also interact the demographics of each zip code with the vehicle’s average price p_{jft} , and I include a disutility of distance d_{zf} between the zip code and dealership that I constrain to be negative using functional form:

$$\mu_{zjft} = x'_j\beta D_z + \alpha'_D D_z p_{jft} - \exp\{\eta_1 + \eta_2 d_{zf}\} \quad (4)$$

Above, x_j, D_z, α_D are column vectors, β is a 4×4 matrix, and η_1, η_2 are scalars.

I normalize the utility of the outside good to be zero, and I assume that the idiosyncratic error ε_{zjft} is IID Type-I extreme value. This means that the probability that a consumer from zip code z chooses vehicle j from dealer f in quarter t can be expressed as

$$P_{zt}[j, f] = \frac{\exp(\delta_{jft} + \mu_{zjft})}{1 + \sum_{f' \in \mathcal{F}} \sum_{j \in \mathcal{J}_{f't}} \exp(\delta_{j'f't} + \mu_{zj'f't})} \quad (5)$$

I calibrate the market size of each zip code using the calculation of Albuquerque and Bronnenberg 2012: The market size M_z is a fixed fraction of the population, $M_z \equiv \lfloor \text{pop}_z / 48 \rfloor$, based on their assumption that a household buys a car once every 6 years and my assumption of two people per household. If we let P_{zt} be the vector collecting the choice probabilities for each possible combination of (j, f) , we can express the demand function implied by this parametrization as

$$D_t(p; \Theta) = \sum_z M_z * P_{zt} \quad (6)$$

Where I collect the parameters of the model into $\Theta \equiv (\xi_f, \xi_j, \xi_{q(t)}, \xi_{y(t)}, \bar{\alpha}, \gamma, \beta, \alpha_D, \eta_1, \eta_2)$

3.1 Estimation

Because I have a full micro-sample for my demand estimation, I take the approach of Goolsbee and Petrin 2004 in maximizing the log-likelihood of my observed purchases over the mean utilities δ and the nonlinear parameters $(\beta_D, \alpha_D, \eta_1, \eta_2) \equiv \theta$, then decomposing δ into its constituent parameters $(\bar{\alpha}, \xi_j, \xi_f, \xi_{q(t)}, \xi_{y(t)}, \gamma) \equiv \chi$. Grieco et al. 2024 note that in my setting, with a just-identified GMM objective, this approach is identical to estimating their more efficient, combined objective.

Recall from the previous section that I defined the vector of choice probabilities for a particular zip code z in quarter t as P_{zt} . I will let C_{zt} denote the vector of observed purchases from z in quarter t . Hence, the log-likelihood of the inside purchases for zt is the dot product

$$\ell_{zt}^{\text{inside}}(\delta, \theta) = C_{zt} \cdot \log P_{zt}(\delta, \theta) \quad (7)$$

Where I use $P_{zt}(\delta, \theta)$ to emphasize that the choice probabilities depend on the current guess of parameters.

In order to calculate the outside share, I write $\mathbf{1} \cdot C_{zt}$ as the total number of cars bought by z in quarter t , hence $(M_z - \mathbf{1} \cdot C_{zt})$ represents the number of potential consumers who did not buy a car. I write P_{0zt} to denote the probability of no purchase, i.e.

$$P_{0zt} = \frac{1}{1 + \sum_{f \in \mathcal{F}} \sum_{j \in \mathcal{J}_{ft}} \exp(\delta_{jft} + \mu_{zjft})} \quad (8)$$

The full log-likelihood of a guess of parameters (δ, θ) is

$$\ell(\delta, \theta) = \sum_t \sum_z [C_{zt} \cdot \log P_{zt}(\delta, \theta) + (M_z - \mathbf{1} \cdot C_{zt}) \log P_{0zt}(\delta, \theta)] \quad (9)$$

I do two things to make this estimation objective numerically stable: I scale the covariates (p_{jft}, D_z, d_{zf}) to have variance one by dividing each variable by its standard deviation; and I scale the objective function ℓ by dividing by the number of observed purchases $N \equiv \sum_{z,t} \mathbf{1} \cdot C_{zt}$. My objective is a function of roughly 65,000 parameters, but after these adjustments, I am able to use a quasi-Newton solver that does not store a full Hessian approximation (the so-called L-BFGS solver of Liu and Nocedal 1989) and find relatively fast convergence given the amenable structure of the likelihood.

After obtaining estimates $(\hat{\delta}, \hat{\theta})$, I estimate the mean parameters χ by running an instrumental variables regression with fixed effects. I use the Hausman instrument to proxy for price p_{jft} and estimate the regression model implied by (3). I report standard errors for parameters θ using the Fisher information matrix, and I report standard errors for the parameters in χ using White’s robust method from the IV regression.

4 Supply

4.1 State Space Reduction and Notion of Equilibrium

I assume that dealers are playing a dynamic game where they are given a sales target and demand conditions as state, and they must choose to set prices in the current period to maximize the expected present value of profits. Dealers know that they must lower their prices to sell a sufficient number of cars to reach their sales target, which in expectation will result in an expanded assortment of cars (and higher profit) in the next period.

The nature of this model makes it difficult to enumerate all of the payoff-relevant features that a dynamic game would need to hold in its state. I have a large number of dealers (218 Ford dealers) who all have

varying degrees of local market power, making atomistic firm assumptions, a common solution to the curse of dimensionality, unreasonable.

The simplification I make to the dynamic game is to treat each firm as a monopolist over its residual demand curve and to treat estimation as a single-agent dynamic problem. It is true by definition that firms are monopolists over their residual demand curves, meaning the single-agent dynamic program that I write below is equivalent to the agent’s program in a dynamic game. The key insight I make is that because Bajari, Benkard, and Levin 2007 is not a full-solution method, and because I do not have to re-compute dynamic equilibrium in my counterfactual estimation, I can leverage the relative simplicity of the single-agent framing to make estimation of this complex dynamic game tractable.

The way that I choose to summarize a dealer’s residual demand is as a system of translog coefficients. I will describe its derivation in detail below—for now, I will note that I use vector ς to denote a dealer’s residual demand state, and the product assortment $\mathcal{J}$ can be encoded jointly as part of state.⁵

Assume that a dealer holds residual demand ς , a sales target φ , and an idiosyncratic private information shock ε as state. A manufacturer sets the sales target for next period according to a function $\phi(\varsigma, \iota)$ —exogenous to the model—that depends on the current period’s state and an indicator ι of whether a dealer met their sales target or not:

$$\iota \equiv 1 \left\{ \sum_j D_j(p; \varsigma, v) \geq \varphi \right\} \quad (10)$$

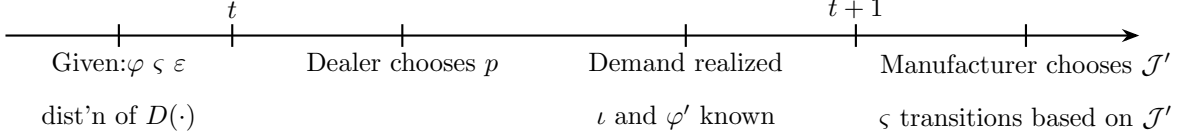
Residual demand $D(\cdot)$ is by definition parametrized by ς , and I also assume that it is stochastic according to random variable v . This is consistent with the dynamic games literature (Aguirregabiria, Collard-Wexler, and Ryan 2021) that treats payoffs as stochastic because other players’ idiosyncratic private information shocks generate probabilistic best responses. It will also become apparent when discussing estimation that residual demand must be stochastic for the model to have coherent timing assumptions.

A dealer with state $(\varsigma, \varphi, \varepsilon)$ chooses prices that solve the Bellman equation

$$V(\varsigma, \varphi, \varepsilon) = \max_{\{p_j | j \in \mathcal{J}\}} \int \left[\sum_{j \in \mathcal{J}} D_j(p; \varsigma, v) [p_j - mc_j - \varepsilon_j] + \beta \int V(\varsigma', \phi(\varsigma, \iota), \varepsilon') dF(\varsigma' | \varsigma, \iota) dF(\varepsilon) \right] dF(v) \quad (11)$$

I will specify a parametrization for marginal cost mc_j later while discussing estimation. State evolution happens as follows:

⁵For the impatient, what I mean by encoding assortment is that my translog coefficients are embedded in a larger sparse vector that has uniform structure across all dealer-quarters. This uniform representation, along with machine learning methods that handle sparsity naturally, makes my estimation feasible.



The dealer knows what its expected residual demand, $p \rightarrow \mathbb{E}_v[D(p; \varsigma, v)]$, will be at the beginning of the period. They then commit to prices, and actual demand is realized afterward. Once demand is realized, we know whether the dealer met their sales target. The indicator ι determines their sales target for the next period, and it also affects the assortment of cars the manufacturer chooses to give the dealer next period. Once the set of cars $\mathcal{J}'$ is determined by the manufacturer, the residual demand state ς transitions stochastically. The dealer must form an expectation both over what products they will have available to them, as well as what their expected residual demand state ς' will be next period, given that set of products available to sell.

Again, I take the manufacturer's actions as exogenous to model, and the goal is only to estimate the manufacturer's policy functions without imposing behavioral assumptions to try and recover manufacturer costs. In the next section I describe how these policies can be identified, and how dealers' unobserved compliance with the sales target can be recovered from the data.

4.2 Compliance and Sales Targets are Identified by Assortment

The core concept to the allocation process is that there is a “base” allocation, depending on state ς , that each dealer gets regardless of performance. Beyond that, there is a sales target $\phi(\varsigma, \iota)$ the manufacturer sets, again taken to be exogenous from the model, and there is a “bonus” allocation that dealers get if they meet their sales target. The fact that firms that are observably similar get the same base allocation, with firms that meet their sales target getting extra trims, is exactly the argument for how the econometrician can identify which firms are meeting their targets and which are not.

I pose the following model for allocations: Suppose that each trim has two functions that map state $\varsigma \rightarrow [0, 1]$ to a particular probability

$$\rho_j^{\text{base}}(\varsigma) \in [0, 1], \rho_j^{\text{bonus}}(\varsigma) \in [0, 1] \quad (12)$$

The probability $\rho_j^{\text{base}}(\varsigma)$ is the probability that, conditional on a dealer's state ς , the manufacturer chooses to include trim j as part of the dealer's base allocation next quarter, regardless of their sales performance.

Conditional on a dealer's state, the manufacturer also chooses a sales target $\phi(\varsigma, \iota)$ for them to meet in quarter t . If a dealer meets their sales target, then they are eligible for additional cars next quarter. This

bonus allocation is governed by $\rho_j^{\text{bonus}}(\varsigma)$: it is the additional probability that, if a dealer meets its sales target, it will be awarded trim j next quarter conditional on state ς .

Again, I use the indicator ι here to mean that a dealer has met their sales target, which I will also call “complying.” Given these two probabilities ρ_j^{base} and ρ_j^{bonus} , we can write the probability of observing a trim j in a dealer’s product set next period as

$$P[j \in \mathcal{J}' \mid \varsigma, \iota] = \begin{cases} \rho_j^{\text{base}}(\varsigma) & \iota = 0 \\ \rho_j^{\text{base}}(\varsigma) + [1 - \rho_j^{\text{base}}(\varsigma)]\rho_j^{\text{bonus}}(\varsigma) & \iota = 1 \end{cases}$$

The allocation model hence implies that the unconditional distribution $P[\mathcal{J}' \mid \varsigma]$ is a mixture of independent Bernoulli draws:

$$P[\mathcal{J}' \mid \varsigma] = \pi \prod_j P[j \in \mathcal{J}' \mid \varsigma, \iota = 1] + (1 - \pi) \prod_j P[j \in \mathcal{J}' \mid \varsigma, \iota = 0] \quad (13)$$

where π is the global probability that dealers meet their sales target. In particular, this is a finite non-parametric mixture. We have strong identification results for nonparametric mixtures due to Allman, Matias, and Rhodes 2009. My modeling choice of assortments as conditionally independent Bernoulli random variables allows me to apply their Theorem 7, which states that as long as the dimension of $\mathcal{J}'$ is greater than two, the nonparametric component probabilities and the mixing probabilities are all separately identified up to a permutation of the component labels. I impose economic structure on my problem so that $P[j \in \mathcal{J}' \mid \varsigma, \iota = 1] \geq P[j \in \mathcal{J}' \mid \varsigma, \iota = 0]$, making my labeling unambiguous as well. Hence, I am able to claim their result that both the component probabilities and the mixing probability are separately identified, and a variation of the standard expectation-maximization procedure from Dempster, Laird, and Rubin 1977 can be used to estimate π and $(\rho^{\text{base}}, \rho^{\text{bonus}})$.

4.3 Estimation

My estimation method proceeds from Bajari, Benkard, and Levin 2007 (henceforth BBL) in the case of a single-agent, continuous action space model. First I must create the state space summary for each dealer and estimate compliance and sales targets offline. Then I estimate reduced-form policy and transition functions, forward-simulate paths for each dealer, and use BBL’s moment inequality estimator to recover dealer costs.

4.3.1 Residual Demand

The state space reduction that I make is to estimate each dealer’s residual demand function at the observed prices and assortments as a translog system. For each dealer, I sample 100 points uniformly from the hypercube $k \in [0.85, 1.15]^{|\mathcal{J}|}$. For each sample b , I use my structural demand system to estimate expected demand $q^{(b)}$ at prices $p^{(b)} = k^{(b)} * p$. Then, I estimate a translog demand system on this sample of points:

$$\ln q_j^{(b)} = \alpha_{0j} + \alpha_{jj} \ln p_j^{(b)} + \sum_{k \neq j} \alpha_{kj} \ln p_j^{(b)} \ln p_k^{(b)} + \varepsilon_{jb} \quad (14)$$

Estimation by standard OLS yields a system of $|\mathcal{J}| * (|\mathcal{J}| + 1)$ coefficients. These coefficients depend on the exact assortment that a dealer has at a given point, so I embed it in a larger sparse matrix that has entries for every possible model year and trim I see in my data.

If I let J represent the number of distinct model year-trim combinations in my data, then each dealer’s state ς can be represented as a $J * (J + 1)$ sparse matrix, where filled rows and columns correspond to the vehicles that a dealer had at a given point in time. In this way ς ’s structure is uniform across dealers, and it encodes both assortment and demand information. I use the XGBoost machine learning model from Chen and Guestrin 2016 to estimate all of the objects that depend on state in my model, which is a tree-based model that handles sparsity naturally as one of its splitting criteria.

4.3.2 Compliance and Sales Targets

I recover the components $(\rho^{\text{base}}, \rho^{\text{bonus}}, \iota, \phi)$ that determine allocations and sales targets using the Expectation-Maximization method from Dempster, Laird, and Rubin 1977 as discussed in Section 4.2. One caveat that applies for nonparametric mixtures and that has been noted in the literature is that there are degenerate solutions. If we let π be our latent parameter of interest, that is, the overall fraction of dealers that met their sales targets, choosing either $\pi = 0$ or $\pi = 1$ will satisfy the convergence criterion of EM and act as an attractor absent any mitigation. Note this is not an identification but rather an estimation issue. Allman, Matias, and Rhodes 2009 gives us identification as $N \rightarrow \infty$, yet the EM algorithm in finite samples is not guaranteed to converge to the identified point.

There are a number of approaches developed in the statistics literature to overcome this issue with poorly behaved EM objectives. One line of work has been to approximate the E and/or M steps using Monte Carlo, hence its name, Monte Carlo Expectation-Maximization (MCEM). Ruth 2024 gives a review of the methodology. I adopt the approach of Wei and Tanner 1990 in sampling the latent variables at each step.

Recall the likelihood of observing a particular trim j in a dealer's product set $\mathcal{J}_{f,t+1}$ given their state ς_{ft} and an indicator ι_{ft} for if they met the sales target in period t :

$$P[j \in \mathcal{J}_{f,t+1} \mid \varsigma_{ft}, \iota_{ft}] = \begin{cases} \rho_j^{\text{base}}(\varsigma_{ft}) & \iota_{ft} = 0 \\ \rho_j^{\text{base}}(\varsigma_{ft}) + [1 - \rho_j^{\text{base}}(\varsigma_{ft})]\rho_j^{\text{bonus}}(\varsigma_{ft}) & \iota_{ft} = 1 \end{cases} \quad (15)$$

I will also assume a statistical model for sales targets that specifies the probability a dealer met their sales target, given observed sales q_{ft} and sales target $\hat{\varphi}$:

$$P[\iota_{ft} = 1 \mid q_{ft}, \hat{\varphi}] = F\left(\frac{q_{ft} - \hat{\varphi}}{\tau}\right) \quad (16)$$

The function F is the logistic CDF,

$$F(x) = \frac{1}{1 + e^{-x}} \quad (17)$$

and τ is a regularization hyperparameter that I use to keep arguments to F in areas with numerically stable density. I calibrate τ to be one-half the standard deviation of q_{ft} in my data.

The EM algorithm repeatedly iterates estimating $(\rho^{\text{base}}, \rho^{\text{bonus}}, \phi)$ and ι in a loop. The algorithm is as follows:

Initialization: We initialize $\iota_{ft}^{(0)} \in \{0, 1\}$ for every dealer-period (f, t) as an indicator for if a dealer f met their sales target in period t . We calculate $\pi^{(0)}$ to be the global average of $\iota^{(0)}$.

M step: Calculate $(\rho^{\text{base}}, \rho^{\text{bonus}})$ to maximize the likelihood of $\mathcal{J}_{f,t+1}$ given guess $\iota_{ft}^{(i)}$. The log-likelihood of observing trim j in dealer f 's product set in period $t + 1$ is

$$\ell_j(\rho_j^{\text{base}}, \rho_j^{\text{bonus}} \mid \varsigma_{ft}, \iota_{ft}^{(i)}, \mathcal{J}_{f,t+1}) = 1\{j \in \mathcal{J}_{f,t+1}\} \ln \left(\rho_j^{\text{base}}(\varsigma_{ft}) + \iota_{ft}^{(i)}(1 - \rho_j^{\text{base}}(\varsigma_{ft}))\rho_j^{\text{bonus}}(\varsigma_{ft}) \right) \quad (18)$$

$$+ 1\{j \notin \mathcal{J}_{f,t+1}\} \ln \left((1 - \iota_{ft}^{(i)})(1 - \rho_j^{\text{base}}(\varsigma_{ft})) + \iota_{ft}^{(i)}(1 - \rho_j^{\text{base}}(\varsigma_{ft}))(1 - \rho_j^{\text{bonus}}(\varsigma_{ft})) \right) \quad (19)$$

Given this likelihood, we can calculate $(\rho_j^{\text{base}}, \rho_j^{\text{bonus}})$ to maximize the probability of the observed dealer assortments $\mathcal{J}_{f,t+1}$. I use alternating maximization with cross-fitting and custom XGBoost objectives, which I describe in Appendix ???. I enforce separation by requiring that $\rho_j^{\text{bonus}} \geq 0.005$ for all j , i.e. that every car has at least a half-percent chance of being allocated as a reward for meeting a sales target. This separation also helps avoid degenerate solutions.

I independently calculate $\hat{\phi}$ by choosing the function $\phi(\varsigma_{ft-1}, \iota_{ft-1})$ to best match dealer sales q_{ft} to the guess $\iota_{ft}^{(i)}$ that they met their sales target. The dealer's sales target next period is a function of their state ς_{ft-1} and an indicator of whether they met their target, $\iota_{ft-1}^{(i)}$. The log-likelihood of a guess of ϕ is

$$\ell(\phi \mid q_{ft}, \iota_{ft}^{(i)}, \varsigma_{ft-1}, \iota_{ft-1}^{(i)}) = \iota_{ft}^{(i)} \ln F \left(\frac{q_{ft} - \phi(\varsigma_{ft-1}, \iota_{ft-1}^{(i)})}{\tau} \right) \quad (20)$$

$$+ (1 - \iota_{ft}^{(i)}) \ln \left(1 - F \left(\frac{q_{ft} - \phi(\varsigma_{ft-1}, \iota_{ft-1}^{(i)})}{\tau} \right) \right) \quad (21)$$

I estimate ϕ using XGBoost with a custom objective function. The number of boosting rounds is determined by a hold-out sample.

E step: Given our estimates $(\hat{\rho}^{\text{base}}, \hat{\rho}^{\text{bonus}}, \hat{\phi})$, we calculate the probability γ that each dealer met their sales target by estimating the combined likelihood under each regime:

$$\gamma_{ft}^{(i+1)} \equiv P[\iota_{ft}^{(i+1)} \mid q_{ft}, \mathcal{J}_{ft+1}, \varsigma_{ft}, \varsigma_{ft-1}, \iota_{ft-1}^{(i)}, \hat{\rho}^{\text{base}}, \hat{\rho}^{\text{bonus}}, \hat{\phi}] = \frac{\pi^{(i)} L_{\hat{\phi}, \hat{\rho}}(\iota_{ft} = 1)}{\pi^{(i)} L_{\hat{\phi}, \hat{\rho}}(\iota_{ft} = 1) + (1 - \pi^{(i)}) L_{\hat{\phi}, \hat{\rho}}(\iota_{ft} = 0)} \quad (22)$$

Where $L_{\hat{\phi}, \hat{\rho}}$ is the combined likelihood of my sales target and assortment probability functions:

$$L_{\hat{\phi}, \hat{\rho}}(\iota_{ft} = k) = \exp \left[\ell(\hat{\phi} \mid q_{ft}, \varsigma_{ft-1}, \iota_{ft-1}^{(i)}, \iota_{ft} = k) + \sum_j \ell_j(\hat{\rho}_j^{\text{base}}, \hat{\rho}_j^{\text{bonus}} \mid \varsigma_{ft}, \mathcal{J}_{ft+1}, \iota_{ft} = k) \right] \quad (23)$$

Note two things about Equation 22. First, probabilities of different trims are independent, as is the sales target estimation, hence I can estimate these separately in parallel and sum. Second, the likelihood is over all trims j sold in period $t + 1$, not just those we observe dealer f selling. I assume that each car sells in a six-quarter period, consistent with my observed data. If a car has model year Y , it is available to be sold from Q4 of $Y - 1$ until Q1 of $Y + 1$.

Loop: For our update I sample $\iota_{ft}^{(i+1)}$ from the computed probability $\gamma_{ft}^{(i+1)}$

$$\iota_{ft}^{(i+1)} \sim \text{Bernoulli}(\gamma_{ft}^{(i+1)}) \quad (24)$$

I sample the update $\pi^{(i+1)}$ from a Beta distribution. I add regularization in the form of a prior weight that prevents $\pi^{(i+1)}$ being drawn from a degenerate distribution. Given my sample size the prior is relatively loose:

$$\pi^{(i+1)} \sim \text{Beta} \left(10 + \sum_{(t,f)} \gamma_{tf}^{(i+1)}, 10 + \sum_{(t,f)} [1 - \gamma_{tf}^{(i+1)}] \right) \quad (25)$$

The termination condition for this algorithm is different than standard EM. Similar to Bayesian Monte Carlo methods, Wei and Tanner 1990 suggest taking a fixed number of samples and discarding after a “burn-in” period. The result is that during the burn-in period, the values of (ι, π) drift towards their fitted values, and then retained samples reflect some of the statistical uncertainty surrounding the estimates $(\hat{\iota}, \hat{\pi})$. From an eye test my estimates converge relatively quickly. I perform 100 iterations of the MCEM loop and drop the first 30 samples.

4.3.3 Policy and Transition Functions

Apart from this latent estimation, I must also estimate policy functions—what prices dealers charge given their residual demand state ς —and state transition, i.e. how ς transitions from one period to the next.

I choose to model price as log-normal and estimate

$$\ln p_j \sim \mathcal{N}(\mu_j(\varsigma, \varphi), \sigma_j^2(\varsigma, \varphi)) \quad (26)$$

for each trim j separately. For every dealer that has j in their product set at period t , I estimate the mean and variance of prices conditional on state ς and sales target φ . In particular, I successively estimate mean and then variance

$$\mu_j(\varsigma, \varphi) = \mathbb{E}[\ln p_j \mid \varsigma, \varphi] \quad (27)$$

$$\sigma_j^2(\varsigma, \varphi) = \mathbb{E}[(\ln p_j - \mu_j(\varsigma, \varphi))^2 \mid \varsigma, \varphi] \quad (28)$$

by fitting XGBoost models to $\ln p_j$ and then its squared residual. I model these price draws across different j as independent—conditioning μ and σ^2 on state ς already encodes information about assortment and demand complementarities, hence I consider this independence assumption mostly innocuous.

The state space is also continuous, so my estimation for state transition proceeds in a similar way. For each product j , I estimate three separate models

$$\alpha_{0j} \sim \mathcal{N}(\mu_{\alpha_{0j}}(\varsigma), \sigma_{\alpha_{0j}}^2(\varsigma)) \quad (29)$$

$$\alpha_{jj} \sim \mathcal{N}(\mu_{\alpha_{jj}}(\varsigma), \sigma_{\alpha_{jj}}^2(\varsigma)) \quad (30)$$

$$\alpha_{kj} \sim \mathcal{N}(\mu_{\alpha_{kj}}(\varsigma, k), \sigma_{\alpha_{kj}}^2(\varsigma, k)) \quad (31)$$

For the dealers that have j in their product set at period t , I estimate the mean and variance α_{0j} based on the state ς in period $t - 1$. I do the same for α_{jj} . For the cross-coefficients α_{kj} , instead of estimating one for each different combination of trims, I save model complexity by including an integer index for the particular trim k so that the XGBoost model can split on it in fitting $(\mu_{\alpha_{kj}}, \sigma_{\alpha_{kj}}^2)$.

4.3.4 Forward Simulation

Calculating cost estimates using BBL requires being able to simulate my model forward. There are two things that make my simulation more complicated than textbook cases: state transition happens conditional on an assortment choice, and single-agent dynamics mask a multi-agent world.

The first issue is that the manufacturer picks a dealer's assortment for next period before state transition happens. This means that state transition is in practice the result of two random draws. First, a manufacturer sees a dealer's state ς and compliance indicator ι , and they choose assortment $\mathcal{J}'$ in the next period as a result of independent Bernoulli draws. The probabilities of each of these draws are calculated according to our fitted functions $(\hat{\rho}^{\text{base}}, \hat{\rho}^{\text{bonus}})$.

Once a random assortment is drawn for next period, then I can calculate state transition. I only draw samples of the coefficients $(\alpha_{0j}, \alpha_{jj}, \alpha_{kj})$ for the (j, k) combinations of vehicles the dealer will have available next period. The rest of the new state matrix ς' will remain empty.

The other complication is the way in which my agents interact with their world. Consider after our state has transitioned and we enter a new quarter. As I have modeled it, dealers choose prices independent of each other, conditional on this new residual demand state. Despite my choice to not explicitly model competitors, residual demand does implicitly depend on the prices that competitors choose. So as I draw a random set of prices for one dealer, another dealer's residual demand changes, implying he must adjust his prices as well. This is an issue of mutual dependence, of prices on state and of state on prices, that just so happens to rear its head in simulating the model forward.

There is a sense in which a fixed point—that is, a price equilibrium—exists, and I could iterate until residual demand and dealer prices are consistent with one another. However, this defeats the purpose of modeling single-agent dynamics and keeping estimation tractable. Hence I make the choice to treat demand as stochastic: After dealers choose their prices, I then *re-calculate* each dealer’s residual demand at the chosen prices and use this for their realized demand as well as the state for calculating the dealer’s sales target and assortment for next period. These are, ultimately, idiosyncratic shocks that drive the differences in residual demand before and after choosing prices, and the state transition function should accurately capture mean residual demand changes, including how pricing pressure from rival dealers may change. As noted above, this is similar to how in the multi-agent context, flow payoffs are stochastic because of competitors’ idiosyncratic private information shocks.

The timing of my BBL simulations is as follows:

1. Starting at $t = 0$, use a dealer’s residual demand ς_{ft} and estimated compliance $\hat{l}_{ft}$ to draw an assortment $\mathcal{J}_{ft+1}$, residual demand ς_{ft+1} , and set sales target $\varphi_{ft+1} \equiv \hat{\phi}(\varsigma_{ft}, \hat{l}_{ft})$. Do this for every dealer f .
2. Given residual demand ς_{ft+1} and sales target φ_{ft+1} , draw dealer prices according to $\ln p_{jft+1} \sim \mathcal{N}(\hat{\mu}_j(\varsigma_{ft+1}, \varphi_{ft+1}), \hat{\sigma}_j^2(\varsigma_{ft+1}, \varphi_{ft+1}))$. Do this for every dealer.
3. After drawing prices, recalculate residual demand according to Section 4.3.1. Call this ς'_{ft+1} . Calculate total demand at prices p_{jft+1} . Call this q_{ft+1} . Do this separately for each dealer.
4. Now we observe $\iota_{ft+1} = 1(q_{ft+1} \geq \varphi_{ft+1})$. Treat ς'_{ft+1} as the new residual demand state. With $(\iota_{ft+1}, \varsigma'_{ft+1})$ in hand, repeat from step 1.
5. Stop at the end of the sample frame.

I stop at the end of the sample frame because I treat different model years as different products j , hence I cannot simulate the same model and trim selection forward into the future. This is a limitation of my modeling choice, and it amounts to a behavioral assumption that dealers have limited foresight.

4.3.5 Cost Estimator

The objects that I aim to recover with my estimation procedure are dealer costs. I assume a simple fixed-effect structure:

$$mc_{jft} = w_j + \zeta_f + \zeta_{q(t)} \tag{32}$$

The object w_j is the wholesale price of a particular trim, which by law is uniform across dealers in Texas. The other fixed effects $\zeta_f, \zeta_{q(t)}$ represent dealer- and quarter-specific cost shocks. The primary object used in estimation is a function $W(\sigma', \varsigma, \varphi)$ that describes the expected value of following policy σ' in state (ς, φ) . (Recall that the firm's policy here is to choose prices as a function of state (ς, φ) , so a strategy σ' represents following a particular pricing strategy.)

$$W(\sigma', \varsigma, \varphi) = \mathbb{E}_{\sigma', v, \varepsilon}[V^{\sigma'}(\varsigma, \varphi, \varepsilon)] \quad (33)$$

As BBL note, because my flow payoff is linear in costs, W is linear in my parameter vector θ

$$W(\sigma', \varsigma, \varphi) = \Phi(\sigma', \varsigma, \varphi) \cdot \theta \quad (34)$$

This means that I can run my forward simulations and pre-compute Φ for every period and strategy, outside of the estimation loop. Hence estimation is incredibly cheap after this initial computation.

One problem is to choose the set of counterfactual strategies to compare against the firm's simulated optimal policy. Thankfully, when I calculate residual demand in the procedure above, I am required to sample perturbations from the simulated optimal price. I simply retain these perturbed prices and the demands they generate during my forward simulations, and these form the counterfactual policies σ' against which to compare my simulated optimal policies.

To be more clear with the bookkeeping, each residual demand calculation computes 150 price perturbations. This gives 150 counterfactual prices and demands. I run 100 forward simulations, meaning that the empirical analog of W is computed with 100 simulation draws in each counterfactual policy. I compute W for every dealer-quarter combination, save the last year. Consistent with the short-sighted behavioral assumption that I make above, I assume that the dealer sets prices considering their realized demand in the current period and a year forward. Hence I use

$$\tilde{W}_{ft}(\sigma', \varsigma, \varphi) \approx \frac{1}{n_S} \sum_{s=1}^{n_S} \sum_{\tau=0}^{t+4} \beta^{\tau-t} \sum_{j \in \mathcal{J}_{f\tau}^{(s)}} D_j(p_{jft}^{(s)}; \varsigma_{ft}^{(s)}, v_{f\tau}^{(s)}) [p_{jft}^{(s)} - mc_{jft}] \quad (35)$$

as my estimate of the dealer's value function in a given state. Superscript (s) denotes simulated values. Prices are drawn according to a particular policy σ'

$$p_{f\tau}^{(s)} \sim \sigma'(\varsigma_{f\tau}^{(s)}, \varphi_{f\tau}^{(s)}) \quad (36)$$

I choose discount factor $\beta = (0.95)^{1/4}$, meaning the dealer discounts at a 5% annual rate. To form my moment inequality estimator, I collect the 150 counterfactual policies per dealer-quarter into a moment function. Let $|ft|$ be the number of unique dealer-quarters I can include (i.e., all save the last year), and let θ be the parameter vector I want to estimate. I define $g : \mathbb{R}^{|\theta|} \rightarrow \mathbb{R}^{150*|ft|}$ as

$$g(\theta)_{\sigma'ft} = \tilde{W}(\sigma^*, \varsigma_{ft}, \varphi_{ft}) - \tilde{W}(\sigma', \varsigma_{ft}, \varphi_{ft}) \quad (37)$$

The $(\sigma', f, t)^{\text{th}}$ entry of g compares the counterfactual pricing strategy σ' to the optimal strategy σ^* for dealer f in quarter t .

My estimate $\hat{\theta}$ is computed using a minimum-distance objective that penalizes violations of the optimal policy. Following the suggestion of Pakes et al. 2015, I scale each moment in g by its standard deviation to balance the informativeness of different moments

$$\hat{\theta} = \arg \min_{\theta} \left\| D(\theta)^{-1/2} * [g(\theta) \odot 1\{g(\theta) < 0\}] \right\| \quad (38)$$

Matrix $D(\theta)$ is a diagonal matrix containing the variance of each of the entries of g across simulations.

Because my marginal cost is parameterized as the sum of fixed effects, I adopt the method of alternating projections to estimate mc_{jft} . This leads to a natural way to express the value function $\tilde{W}$ for $(w_j, \zeta_f, \zeta_{q(t)})$. Each fixed effect has its own Φ matrix with the same number of columns as there are levels of the fixed effect. Starting with an initial guess $(w_j^{(0)}, \zeta_f^{(0)}, \zeta_{q(t)}^{(0)})$, I alternate estimating $(w_j, \zeta_f, \zeta_{q(t)})$ using $\Phi^{(j)}$, $\Phi^{(f)}$, and $\Phi^{(t)}$, respectively, until my estimates converge.

5 Results

Table 4 contains coefficient estimates for my demand model. Because all covariates are standardized, the estimates in the top panel of the table are directly comparable. How urban a zip code is tends to dominate other demographic effects, with more urban zip codes tending to have lower baseline preference for each vehicle type, as well as higher price sensitivity. (This is *ceteris paribus*; urban zip codes also tend to be higher income, whose coefficient estimate implies lower price sensitivity.) I calculate the median own-price elasticity across all of the vehicles offered as roughly 6.8. This implies that median (static) markups should be on the order of 15%, a large departure from the estimate of Brunson and Barnes 2020 that markups are 2–3% above wholesale cost. This difference may be attributable to the dynamic incentives of allocation, but it may also be due to dealers internalizing the lifetime service revenue a consumer generates.

Table 4: Demand Model, Estimated Coefficients

	Inv. Median Earnings	Frac. Urban	Frac. Hispanic	Frac. Black
Coupe	0.0329*** (0.0137)	−0.196*** (0.0140)	0.0151 (0.0150)	−0.0224*** (0.0110)
SUV	0.159*** (0.00897)	−0.353*** (0.00861)	−0.141*** (0.0101)	−0.0901*** (0.00797)
Pickup	0.168*** (0.0117)	−0.399*** (0.0109)	−0.0564*** (0.0129)	−0.110*** (0.0103)
Sedan	0.199*** (0.00784)	−0.345*** (0.00858)	−0.0247*** (0.00921)	−0.0488*** (0.00699)
Price	−0.0655*** (0.00393)	−0.0539*** (0.00358)	−0.0424*** (0.00352)	−0.0617*** (0.00434)
Mean Price Effect	−3.203*** (0.174)	η_1 (Dist. Baseline)		2.400*** (0.00373)
$\log(J)$	−1.025** (0.0289)	η_2 (Distance)		0.339*** (9.46×10^{-4})
Median Own-Price Elasticity	−6.848			

In order to interpret the magnitude of the distance coefficients, I calculate a willingness to pay for distance metric. For each observed purchase in my data, I calculate the indirect utility from the purchase, then calculate what change in price would have made the purchaser indifferent between their actual choice and the same choice but one mile closer. Figure 1 plots the histogram of this willingness to pay for proximity. Because I do not model any heterogeneity in the distance utility, dispersion in these estimates is caused by dispersion in the price coefficient, as well as from the curvature of the utility term $-\exp\{\eta_1 + \eta_2 d_{zf}\}$ itself.

Figure 1 shows a fairly large average willingness to pay for distance, with the median estimate being about \$130 per mile. This is likely driven by the fact that there are two types of consumers in my data: Roughly half of consumers do not travel, rather opting to buy from the dealership closest to them. The other half travel an average of about 40 miles to purchase their car, with a significant portion traveling over 100 miles. The homogeneity of the distance disutility term hence fails to differentiate between these two types of consumers, nor does it capture the relationship between price coefficient heterogeneity and distance disutility heterogeneity. (For instance, higher income consumers may travel less because they are less concerned about price, above and beyond what is implied by their price coefficient.) Subsequent demand modeling should capture this two-type demand model in the same way that airline papers like Williams 2022 model latent “business” and “leisure” consumers.

Figure 2 shows a scatterplot of total dealer sales, q_{ft} , against their estimated sales targets, φ_{ft} . The color of the points represents the inferred probability that each dealer met their sales target. The plot shows that

Figure 1: Distribution of WTP for Proximity

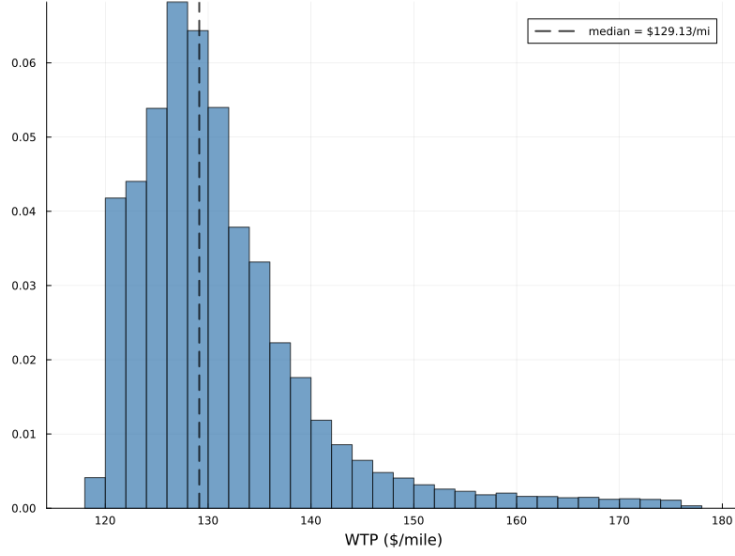
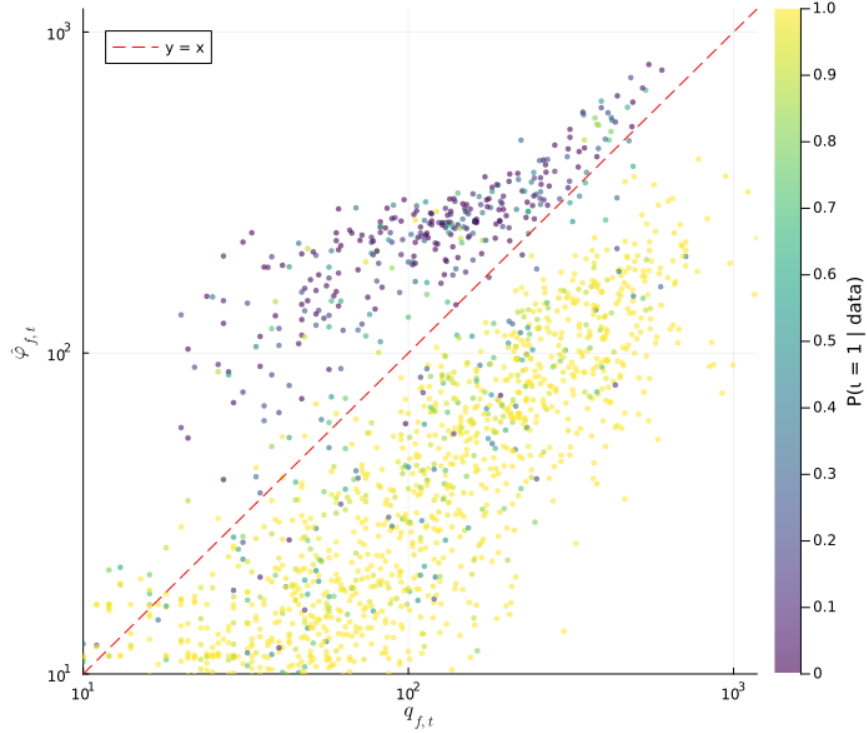


Figure 2: Sales Targets and Compliance Probabilities



the estimation procedure separates dealers into two groups with modest success. Compliers are not simply dealers that have higher sales, but the joint estimation of assortments with sales targets separates complying and non-complying dealers along the entire spectrum of low- to high-performing dealers.

Figure 3: Select Trim Probabilities

Top 6 popular (rows 1-2) vs. mid-popularity, top std($\hat{p}$) (rows 3-4)

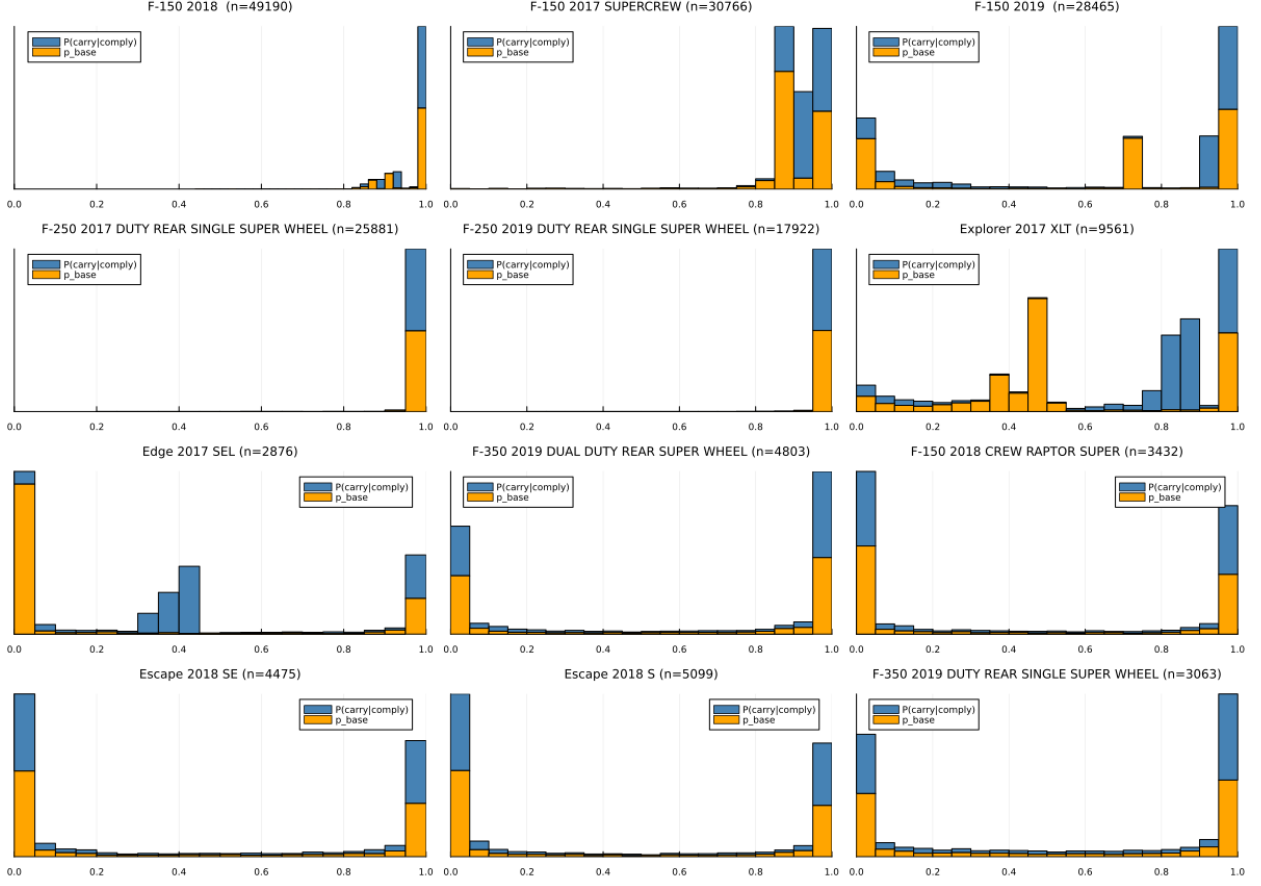
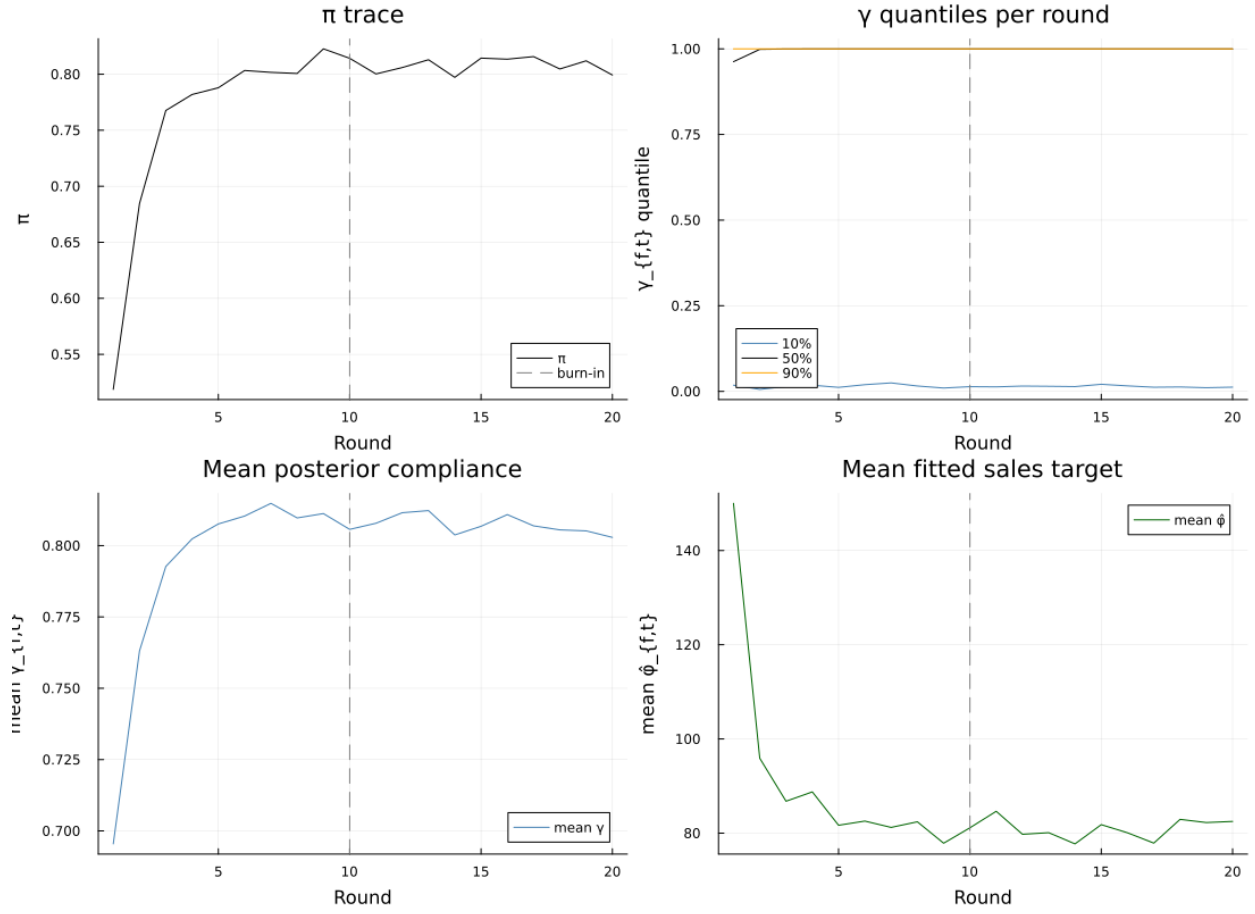


Figure 3 shows examples of assortment probabilities for different trims, as well as how these trims can be used in different ways as either a “necessity” for most dealerships or a reward for dealer performance. Each subplot is a stacked histogram. The orange histogram shows the base probability across all dealers that they are awarded the trim, without the dealer meeting the sales target. The blue histogram calculates the probability that the dealer is awarded the trim if they meet the sales target. A large rightward shift from orange to blue indicates trims that are estimated to act as a strong reward. (Put another way, these are trims the estimation algorithm has identified as a strong differentiator between complying and non-complying dealers.) Some trims, like the base Ford F-150 variants, do not show a strong shift from orange to blue, consistent with the fact that most dealers are allocated these cars whether or not they meet their sales targets. Appendix Table ?? contains a complete listing of base and bonus probabilities for every trim in my data.

Figure 4: MCEM Traces



Finally, Figure 4 shows the trace of the compliance estimate across iterations. The global estimate of π converges quickly to just over 80% and remains stable throughout. This is despite the re-sampling of latent variables at each MCEM iteration. This estimate informs my finding that sales targets are broadly achievable, with about 4 of every 5 dealers meeting the sales target set for them.

References

- Akerberg, Daniel A. and Marc Rysman (2005). “Unobserved Product Differentiation in Discrete-Choice Models: Estimating Price Elasticities and Welfare Effects”. In: *The RAND Journal of Economics* 36.4, pp. 771–788.
- Aguirregabiria, Victor, Allan Collard-Wexler, and Stephen P. Ryan (Jan. 2021). “Chapter 4 - Dynamic games in empirical industrial organization”. In: *Handbook of Industrial Organization*. Ed. by Kate Ho, Ali Hortaçsu, and Alessandro Lizzeri. Vol. 4. Handbook of Industrial Organization, Volume 4. Elsevier, pp. 225–343.
- Albuquerque, Paulo and Bart J. Bronnenberg (Jan. 2012). “Measuring the Impact of Negative Demand Shocks on Car Dealer Networks”. In: *Marketing Science* 31.1, pp. 4–23.
- Allman, Elizabeth S., Catherine Matias, and John A. Rhodes (Dec. 2009). “Identifiability of parameters in latent structure models with many observed variables”. en. In: *The Annals of Statistics* 37.6A, pp. 3099–3132.
- Bajari, Patrick, C. Lanier Benkard, and Jonathan Levin (2007). “Estimating Dynamic Models of Imperfect Competition”. en. In: *Econometrica* 75.5. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1468-0262.2007.00796.x>, pp. 1331–1370.
- Biesebroeck, Johannes Van and Frank Verboven (2024). “Demand, Competition and Public Policy in the Automobile Industry”. In.
- Brunson, Kendrick and Steve Barnes (2020). “Automotive Dealership Management Fundamentals”. In: *Liberty University Faculty Presentations and Publications*.
- Chen, Tianqi and Carlos Guestrin (Aug. 2016). “XGBoost: A Scalable Tree Boosting System”. In: arXiv:1603.02754 [cs.LG], pp. 785–794.
- Conlon, Christopher T and Julie Holland Mortimer (2021). “Efficiency and Foreclosure Effects of Vertical Rebates: Empirical Evidence”. en. In.
- Dempster, A. P., N. M. Laird, and D. B. Rubin (Sept. 1977). “Maximum Likelihood from Incomplete Data Via the *EM* Algorithm”. en. In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 39.1, pp. 1–22.
- Goolsbee, Austan and Amil Petrin (2004). “The Consumer Gains from Direct Broadcast Satellites and the Competition with Cable TV”. en. In: *Econometrica* 72.2. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1468-0262.2004.00494.x>, pp. 351–381.
- Grieco, Paul L E et al. (2024). *Conformant and Efficient Estimation of Discrete Choice Demand Models*. en.

- Jindal, Pranav and Peter Newberry (Feb. 2022). “The Profitability of Revenue-Based Quotas Under Price Negotiation”. en. In: *Management Science* 68.2, pp. 917–940.
- Lafontaine, Francine and Fiona Scott Morton (Sept. 2010). “Markets: State Franchise Laws, Dealer Terminations, and the Auto Crisis”. In: *Journal of Economic Perspectives* 24.3, pp. 233–250.
- Lafontaine, Francine and Margaret Slade (July 2007). “Vertical Integration and Firm Boundaries: The Evidence”. en. In: *Journal of Economic Literature* 45.3, pp. 629–685.
- Lafontaine, Francine and Margaret E. Slade (May 1998). *Incentive Contracting and the Franchise Decision*. Working Paper. National Bureau of Economic Research: [6544](#).
- Lee, Robin S., Michael D. Whinston, and Ali Yurukoglu (Jan. 2021). “Chapter 9 - Structural Empirical Analysis of Contracting in Vertical Markets”. In: *Handbook of Industrial Organization*. Ed. by Kate Ho, Ali Hortaçsu, and Alessandro Lizzeri. Vol. 4. Handbook of Industrial Organization, Volume 4. Elsevier, pp. 673–742.
- Liu, Dong C. and Jorge Nocedal (Aug. 1989). “On the limited memory BFGS method for large scale optimization”. en. In: *Mathematical Programming* 45.1, pp. 503–528.
- Misra, Sanjog and Harikesh S. Nair (Sept. 2011). “A structural model of sales-force compensation dynamics: Estimation and field implementation”. en. In: *Quantitative Marketing and Economics* 9.3, pp. 211–257.
- Murry, Charles (Jan. 2017). *Advertising in Vertical Relationships: An Equilibrium Model of the Automobile Industry*. en. SSRN Scholarly Paper. Rochester, NY.
- Nurski, Laura and Frank Verboven (July 2016). “Exclusive Dealing as a Barrier to Entry? Evidence from Automobiles”. In: *The Review of Economic Studies* 83.3, pp. 1156–1188.
- Pakes, A. et al. (2015). “Moment Inequalities and Their Application”. en. In: *Econometrica* 83.1. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA6865>, pp. 315–334.
- Rey, Patrick and Jean Tirole (1986). “The Logic of Vertical Restraints”. In: *The American economic review*, pp. 921–939.
- Ruth, William (Jan. 2024). *A review of Monte Carlo-based versions of the EM algorithm*. en. arXiv:2401.00945 [stat.CO].
- Smith, Richard L. (Apr. 1982). “Franchise Regulation: An Economic Analysis of State Restrictions on Automobile Distribution”. In: *The Journal of Law and Economics* 25.1, pp. 125–157.
- Tirole, Jean (1988). *The theory of industrial organization*. Cambridge, Mass: MIT Press.
- Wei, Greg C. G. and Martin A. Tanner (Sept. 1990). “A Monte Carlo Implementation of the EM Algorithm and the Poor Man’s Data Augmentation Algorithms”. en. In: *Journal of the American Statistical Association* 85.411, pp. 699–704.

Williams, Kevin R. (2022). “The Welfare Effects of Dynamic Pricing: Evidence From Airline Markets”. en.
In: *Econometrica* 90.2, pp. 831–858.

A Full List of Trim Base and Bonus Probabilities

Table 5: Caption

Model	Year	Trim	n	$\bar{\rho}^{\text{base}}$	$\sigma(\rho^{\text{base}})$	$\bar{\rho}^{\text{bonus}}$	$\sigma(\rho^{\text{bonus}})$	$\bar{P}(j \in \mathcal{J} \mid \iota = 1)$
Edge	2017	SE	1134	0.140	0.326	0.112	0.255	0.164
Edge	2017	SEL	2876	0.227	0.395	0.462	0.270	0.502
Edge	2017	SPORT	246	0.025	0.127	0.036	0.137	0.047
Edge	2017	TITANIUM	2202	0.156	0.340	0.169	0.332	0.198
Edge	2018	SE	1204	0.177	0.350	0.164	0.314	0.220
Edge	2018	SEL	2718	0.281	0.419	0.258	0.380	0.327
Edge	2018	SPORT	357	0.038	0.158	0.057	0.184	0.070
Edge	2018	TITANIUM	1893	0.211	0.382	0.174	0.319	0.244
Edge	2019	SE	496	0.105	0.255	0.069	0.174	0.135
Edge	2019	SEL	2957	0.322	0.442	0.212	0.336	0.339
Edge	2019	ST	586	0.097	0.265	0.153	0.312	0.170
Edge	2019	TITANIUM	1844	0.267	0.405	0.187	0.302	0.299
Escape	2017	S	4790	0.189	0.378	0.158	0.325	0.202
Escape	2017	SE	5533	0.193	0.381	0.179	0.338	0.215
Escape	2017	TITANIUM	4302	0.177	0.367	0.152	0.311	0.198
Escape	2018	S	5099	0.404	0.447	0.349	0.407	0.451
Escape	2018	SE	4475	0.411	0.448	0.356	0.406	0.459
Escape	2018	SEL	2612	0.281	0.411	0.264	0.381	0.331
Escape	2018	TITANIUM	1377	0.182	0.348	0.176	0.318	0.232
Escape	2019	S	2923	0.293	0.431	0.233	0.360	0.316
Escape	2019	SE	3433	0.295	0.431	0.239	0.361	0.322
Escape	2019	SEL	1754	0.230	0.394	0.185	0.319	0.260
Escape	2019	TITANIUM	985	0.144	0.323	0.142	0.281	0.187
Expedition	2017	KING RANCH XLT	2809	0.214	0.381	0.216	0.342	0.263

Continued on next page

Table 5: Caption (Continued)

Model	Year	Trim	n	$\bar{\rho}^{\text{base}}$	$\sigma(\rho^{\text{base}})$	$\bar{\rho}^{\text{bonus}}$	$\sigma(\rho^{\text{bonus}})$	$\bar{P}(j \in \mathcal{J} \mid \iota = 1)$
Expedition	2017	LIMITED	932	0.086	0.252	0.080	0.204	0.117
Expedition	2017	PLATINUM	309	0.024	0.123	0.026	0.100	0.040
Expedition	2018	LIMITED	1397	0.215	0.375	0.166	0.310	0.246
Expedition	2018	PLATINUM	525	0.086	0.247	0.062	0.170	0.112
Expedition	2018	XLT	1719	0.244	0.391	0.196	0.327	0.283
Expedition	2019	LIMITED	1352	0.183	0.352	0.154	0.283	0.228
Expedition	2019	PLATINUM	638	0.086	0.240	0.090	0.224	0.125
Expedition	2019	XLT	2475	0.278	0.422	0.250	0.372	0.314
Explorer	2017	BASE	2621	0.168	0.356	0.147	0.308	0.192
Explorer	2017	LIMITED	3853	0.199	0.376	0.203	0.339	0.246
Explorer	2017	PLATINUM	1123	0.091	0.262	0.096	0.243	0.124
Explorer	2017	SPORT	1507	0.113	0.296	0.120	0.278	0.150
Explorer	2017	XLT	9561	0.549	0.293	0.657	0.270	0.792
Explorer	2018	BASE	1737	0.177	0.356	0.174	0.329	0.214
Explorer	2018	LIMITED	3135	0.267	0.415	0.227	0.368	0.296
Explorer	2018	PLATINUM	786	0.093	0.264	0.091	0.234	0.122
Explorer	2018	SPORT	1181	0.144	0.323	0.132	0.286	0.176
Explorer	2018	XLT	7309	0.377	0.457	0.314	0.408	0.403
Explorer	2019	BASE	408	0.062	0.207	0.049	0.158	0.083
Explorer	2019	LIMITED	867	0.121	0.292	0.119	0.263	0.161
Explorer	2019	PLATINUM	206	0.027	0.136	0.018	0.072	0.037
Explorer	2019	SPORT	487	0.070	0.219	0.084	0.217	0.110
Explorer	2019	XLT	3812	0.334	0.441	0.265	0.373	0.364
F-150	2017	CAB REGULAR	1666	0.153	0.339	0.113	0.267	0.171
F-150	2017	RAPTOR SUPERCREW	2012	0.102	0.279	0.067	0.202	0.115
F-150	2017	SUPERCAB	3513	0.224	0.393	0.209	0.336	0.265

Continued on next page

Table 5: Caption (Continued)

Model	Year	Trim	n	$\bar{\rho}^{\text{base}}$	$\sigma(\rho^{\text{base}})$	$\bar{\rho}^{\text{bonus}}$	$\sigma(\rho^{\text{bonus}})$	$\bar{P}(j \in \mathcal{J} \mid \iota = 1)$
F-150	2017	SUPERCREW	30766	0.914	0.099	0.419	0.368	0.929
F-150	2018		49190	0.970	0.051	0.625	0.323	0.974
F-150	2018	CREW RAPTOR SUPER	3432	0.423	0.457	0.387	0.411	0.483
F-150	2019		28465	0.612	0.411	0.633	0.361	0.718
F-150	2019	CREW RAPTOR SUPER	2347	0.277	0.412	0.224	0.354	0.309
F-250	2017	SUPER DUTY SINGLE	25881	0.971	0.038	0.519	0.316	0.980
F-250	2018	SUPER DUTY SINGLE	8872	0.333	0.449	0.268	0.389	0.350
F-250	2019	SUPER DUTY SINGLE	17922	0.990	0.024	0.618	0.293	0.994
F-350	2017	SUPER DUTY DOUBLE	5424	0.312	0.433	0.267	0.386	0.338
F-350	2017	SUPER DUTY SINGLE	4185	0.294	0.427	0.262	0.382	0.325
F-350	2018	SUPER DUTY DOUBLE	2019	0.185	0.363	0.157	0.309	0.214
F-350	2018	SUPER DUTY SINGLE	1846	0.198	0.372	0.155	0.312	0.225
F-350	2019	SUPER DUTY DOUBLE	4803	0.560	0.451	0.424	0.367	0.598
F-350	2019	SUPER DUTY SINGLE	3063	0.541	0.426	0.339	0.348	0.571
Fiesta	2017	S	570	0.087	0.250	0.077	0.204	0.116
Fiesta	2017	SE	2140	0.226	0.390	0.201	0.350	0.256
Fiesta	2017	ST	166	0.014	0.086	0.028	0.112	0.036
Fiesta	2018	S	567	0.082	0.242	0.055	0.159	0.105
Fiesta	2018	SE	2127	0.222	0.386	0.194	0.319	0.266
Fiesta	2018	ST	47	0.002	0.027	0.010	0.047	0.011
Fiesta	2019	S	872	0.134	0.307	0.098	0.212	0.166
Fiesta	2019	SE	2647	0.259	0.414	0.213	0.360	0.284
Fiesta	2019	ST	220	0.029	0.134	0.032	0.118	0.049
Focus	2017	RS	305	0.037	0.162	0.041	0.142	0.059
Focus	2017	S	1724	0.142	0.332	0.110	0.264	0.157
Focus	2017	SE	3462	0.205	0.386	0.160	0.315	0.220

Continued on next page

Table 5: Caption (Continued)

Model	Year	Trim	n	$\bar{\rho}^{\text{base}}$	$\sigma(\rho^{\text{base}})$	$\bar{\rho}^{\text{bonus}}$	$\sigma(\rho^{\text{bonus}})$	$\bar{P}(j \in \mathcal{J} \mid \iota = 1)$
Focus	2017	SEL	1437	0.118	0.303	0.121	0.282	0.149
Focus	2017	ST	833	0.086	0.261	0.103	0.269	0.118
Focus	2017	TITANIUM	507	0.064	0.221	0.061	0.192	0.088
Focus	2018	RS	53	0.002	0.030	0.010	0.042	0.012
Focus	2018	S	1562	0.138	0.323	0.107	0.251	0.162
Focus	2018	SE	3085	0.267	0.415	0.203	0.335	0.295
Focus	2018	SEL	1079	0.115	0.297	0.085	0.225	0.137
Focus	2018	ST	687	0.077	0.244	0.061	0.184	0.096
Focus	2018	TITANIUM	379	0.048	0.184	0.051	0.161	0.076
Fusion	2017	HYBRID SE	129	0.001	0.014	0.010	0.045	0.011
Fusion	2017	S	1463	0.110	0.291	0.069	0.189	0.124
Fusion	2017	SE	5619	0.202	0.385	0.219	0.306	0.272
Fusion	2017	SPORT	206	0.022	0.117	0.019	0.076	0.035
Fusion	2017	TITANIUM	555	0.047	0.179	0.065	0.195	0.080
Fusion	2018	FWD SE	4165	0.320	0.430	0.263	0.372	0.363
Fusion	2018	HYBRID SE	364	0.052	0.195	0.037	0.137	0.068
Fusion	2018	S	1571	0.183	0.356	0.150	0.288	0.225
Fusion	2018	SPORT	56	0.002	0.029	0.011	0.048	0.013
Fusion	2019	FWD SE	3391	0.312	0.431	0.248	0.352	0.352
Fusion	2019	FWD SEL	214	0.035	0.148	0.044	0.137	0.063
Fusion	2019	HYBRID SE	73	0.003	0.032	0.012	0.053	0.014
Fusion	2019	S	1316	0.182	0.356	0.133	0.269	0.206
Fusion	2019	SPORT	17	0.000	0.000	0.006	0.018	0.007
Mustang	2017	COUPE GT	3871	0.173	0.361	0.170	0.334	0.201
Mustang	2017	COUPE I4	2340	0.140	0.328	0.149	0.319	0.173
Mustang	2017	COUPE SHELBY	448	0.052	0.195	0.067	0.181	0.086

Continued on next page

Table 5: Caption (Continued)

Model	Year	Trim	n	$\bar{\rho}^{\text{base}}$	$\sigma(\rho^{\text{base}})$	$\bar{\rho}^{\text{bonus}}$	$\sigma(\rho^{\text{bonus}})$	$\bar{P}(j \in \mathcal{J} \mid \iota = 1)$
Mustang	2017	COUPE V6	2337	0.148	0.336	0.130	0.285	0.174
Mustang	2018	COUPE GT	2860	0.208	0.378	0.182	0.328	0.243
Mustang	2018	COUPE I4	3523	0.240	0.395	0.220	0.354	0.282
Mustang	2018	COUPE SHELBY	307	0.047	0.182	0.052	0.158	0.076
Mustang	2019	BULLITT COUPE	362	0.070	0.209	0.065	0.170	0.106
Mustang	2019	COUPE GT	2901	0.295	0.427	0.228	0.342	0.324
Mustang	2019	COUPE I4	2848	0.287	0.421	0.276	0.385	0.339
Mustang	2019	COUPE SHELBY	166	0.028	0.122	0.041	0.128	0.057